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# **Reinforcement Learning -RL-**

## **Deep Reinforcement Learning -Proximal Policy Optimization-**

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# Proximal Policy Optimization

- **The data distributions of our observations and rewards are constantly changing as our agent learns.**
  - This is a source of instability in the whole training process.
- **Hyperparameters are important.**
  - Imagine that with a very high learning rate we update the current policy. This causes the next data collection for training to be carried out under a very poor policy.
- **PPO introduces the following:**
  - We want to make the training of the policy more stable by limiting the changes you make to the policy each time you train. This helps to avoid making too big a change to the policy.

# Proximal Policy Optimization

- We know from experience that when there are smaller policy updates during training, the updates are more likely to lead to the best possible solution.
- With PPO, we update the policy in a careful way. To do this, we need to compare how much the current policy has changed compared to the previous one. We can do this by using a ratio calculation between the current and previous policy.
- We then limit this range. This means that we remove the motivation for the current policy to differ too much from the old one..

$$[1 - \epsilon, 1 + \epsilon]$$

# Proximal Policy Optimization

- REINFORCE
  - The objective function

$$L^{PG}(\theta) = E_t[\log \pi_{\theta}(a_t|s_t) * A_t]$$

- The update rule

$$\theta \leftarrow \theta + \alpha \gamma^t (g_t^m - b(s_t^m)) \nabla \ln(a_t^m | s_t^m, \theta)$$

- Here we have a problem
  - If the step size is too small, the training is too slow.
  - If the step size is too high, there is too much variability in the training.

# Proximal Policy Optimization

- Clip the policy:
  - This means that the updated policy is not too far away from the previous policy.  $L^{CLIP}(\theta) = \hat{E}_t[\min(\underbrace{r_t(\theta)\hat{A}_t}_{\text{Normal policy Gradients objective}}, \underbrace{\text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon)\hat{A}_t}_{\text{Clipped version of Normal policy Gradients objective}})]$
  - The ratio function  $r_t(\theta)$ :
    - It is the probability of taking action  $a_t$  at state  $s_t$  in the current policy, divided by the same for the previous policy.
      - » If  $r_t(\theta) > 1$ , the action  $a_t$  at state  $s_t$  is more likely in the current policy than the old policy.
      - » If  $0 < r_t(\theta) < 1$ , the action  $a_t$  is less likely for the current policy than for the old one.

$$r_t(\theta) = \frac{\pi_{\theta}(a_t|s_t)}{\pi_{\theta_{old}}(a_t|s_t)}$$

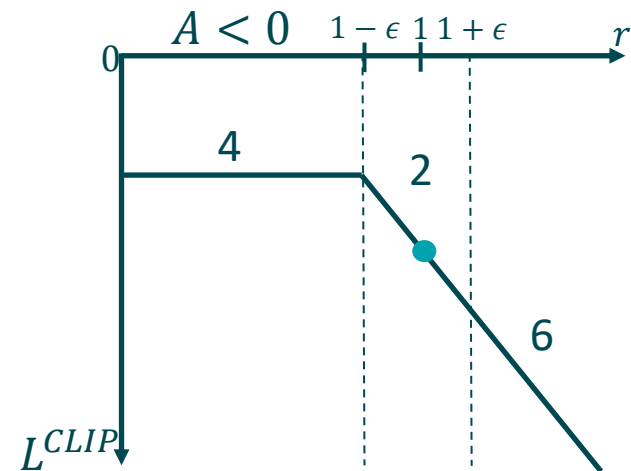
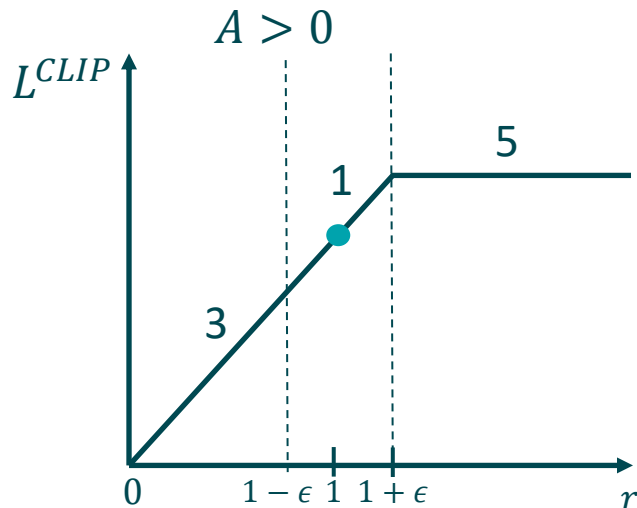
Without a constraint is the action taken is much more probable in our current policy than in our former. This would lead to a significant policy gradient step.

# Proximal Policy Optimization

- Clip the policy:

$r_t(\theta)$

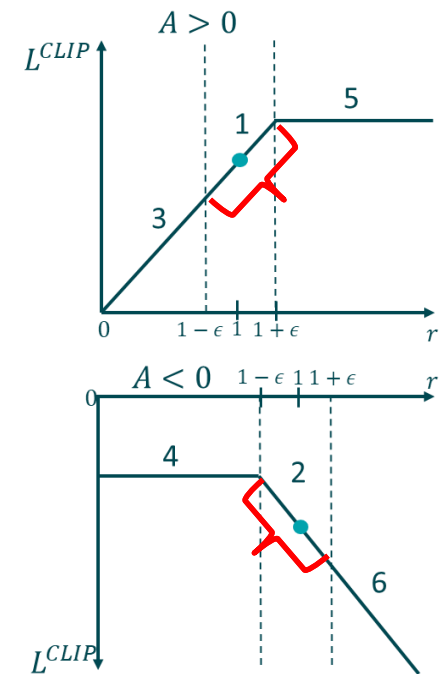
|   | $p_t(\theta) > 0$                              | $A_t$ | Return Value of $\min$ | Objective is Clipped | Sign of Objective | Gradient |
|---|------------------------------------------------|-------|------------------------|----------------------|-------------------|----------|
| 1 | $p_t(\theta) \in [1 - \epsilon, 1 + \epsilon]$ | +     | $p_t(\theta)A_t$       | no                   | +                 | ✓        |
| 2 | $p_t(\theta) \in [1 - \epsilon, 1 + \epsilon]$ | -     | $p_t(\theta)A_t$       | no                   | -                 | ✓        |
| 3 | $p_t(\theta) < 1 - \epsilon$                   | +     | $p_t(\theta)A_t$       | no                   | +                 | ✓        |
| 4 | $p_t(\theta) < 1 - \epsilon$                   | -     | $(1 - \epsilon)A_t$    | yes                  | -                 | 0        |
| 5 | $p_t(\theta) > 1 + \epsilon$                   | +     | $(1 + \epsilon)A_t$    | yes                  | +                 | 0        |
| 6 | $p_t(\theta) > 1 + \epsilon$                   | -     | $p_t(\theta)A_t$       | no                   | -                 | ✓        |



# Proximal Policy Optimization

## • Case 1 and 2:

- The clipping does not apply since the ratio is between the range  $[1 - \epsilon, 1 + \epsilon]$
- In **situation 1**, we have an advantage: the action is better than the average of all the actions in that state. So, we should encourage our current policy to increase the probability of taking that action in that state.
- In **situation 2**, we have a negative advantage. This means that the action is worse than the average of all actions at that state. So, we shouldn't keep doing that action in that state.

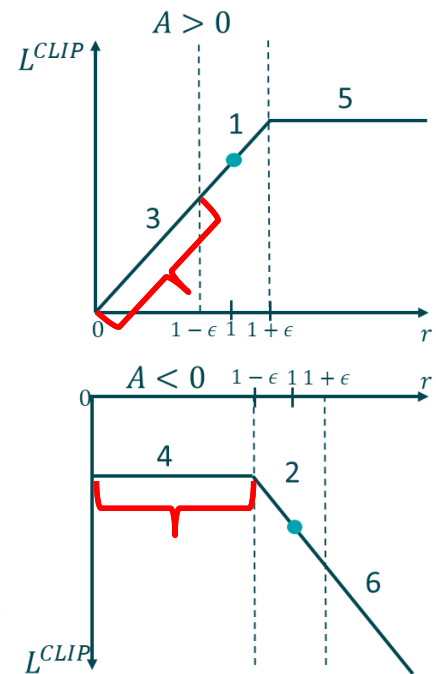


|   | $p_t(\theta) > 0$                              | $A_t$ | Return Value of $\min$ | Objective is Clipped | Sign of Objective | Gradient |
|---|------------------------------------------------|-------|------------------------|----------------------|-------------------|----------|
| 1 | $p_t(\theta) \in [1 - \epsilon, 1 + \epsilon]$ | +     | $p_t(\theta) A_t$      | no                   | +                 | ✓        |
| 2 | $p_t(\theta) \in [1 - \epsilon, 1 + \epsilon]$ | -     | $p_t(\theta) A_t$      | no                   | -                 | ✓        |

# Proximal Policy Optimization

## • Case 3 and 4

- In **situation 3**, the advantage estimate is positive ( $A > 0$ ). This means that you want to increase the probability of taking that action in that state.
- In **situation 4**, the advantage estimate is negative. We don't want to make it less likely that we will take that action in that situation. So, the gradient is equal to zero (because we're on a flat line), which means we don't update our weights.



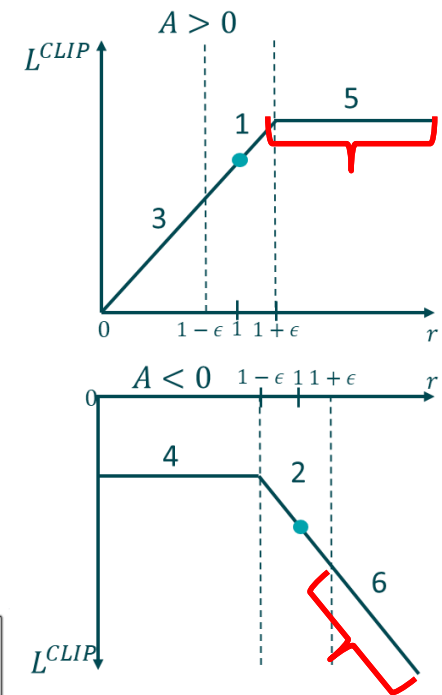
|   | $p_t(\theta) > 0$            | $A_t$ | Return Value of $\min$ | Objective is Clipped | Sign of Objective | Gradient |
|---|------------------------------|-------|------------------------|----------------------|-------------------|----------|
| 3 | $p_t(\theta) < 1 - \epsilon$ | +     | $p_t(\theta)A_t$       | no                   | +                 | ✓        |
| 4 | $p_t(\theta) < 1 - \epsilon$ | -     | $(1 - \epsilon)A_t$    | yes                  | -                 | 0        |



# Proximal Policy Optimization

## • Case 5 and 6

- In **situation 5**, the advantage is positive, so we shouldn't be too greedy. We're already more likely to take that action in that state than in the previous policy. Because of this, the gradient is equal to zero (since we're on a flat line), so we don't update our weights.
- In **situation 6**, the advantage is negative. We want to lower the probability of taking that action in that situation.



|   | $p_t(\theta) > 0$            | $A_t$ | Return Value of $\min$ | Objective is Clipped | Sign of Objective | Gradient |
|---|------------------------------|-------|------------------------|----------------------|-------------------|----------|
| 5 | $p_t(\theta) > 1 + \epsilon$ | +     | $(1 + \epsilon)A_t$    | yes                  | +                 | 0        |
| 6 | $p_t(\theta) > 1 + \epsilon$ | -     | $p_t(\theta)A_t$       | no                   | -                 | ✓        |

# Proximal Policy Optimization

- **We update our policy only if:**
  - Our ratio is in the range  $[1 - \epsilon, 1 + \epsilon]$
  - Our ratio is outside the range, but the advantage leads to getting closer to the range
    - Being below the ratio but the advantage is  $> 0$
    - Being above the ratio but the advantage is  $< 0$

# Proximal Policy Optimization

- The final PPO objective:

$$L^{CLIP}(\theta) = \hat{E}_t[\min(r_t(\theta)\hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon)\hat{A}_t)]$$

$$L_t^{PPO}(\theta) = \hat{E}_t[L_t^{CLIP}(\theta) - \underbrace{c_1 L_t^{VF}(\theta)}_{\text{Value error term}} + \underbrace{c_2 S[\pi_\theta](s_t)}_{\text{Entropy term}}]$$

- And the final update rule:

$$\theta \leftarrow \theta + \alpha \nabla_\theta L_t^{PPO}(\theta)$$

# Proximal Policy Optimization

- The value error term

$$L_t^{PPO}(\theta) = \hat{E}_t[L_t^{CLIP}(\theta) - c_1 L_t^{VF}(\theta) + c_2 S[\pi_\theta](s_t)]$$

$$L_t^{VF}(\theta) = (V_\theta(s_t) - V_t^{targ})^2$$

- The  $c_1$  parameter determines how much importance the algorithm places on minimizing the value function error compared to optimizing the policy.
- If  $c_1$  is too large, the algorithm may focus too much on learning the value function instead of improving the policy.
- If  $c_1$  is too small, the value function may not be learned correctly. This can lead to inaccurate advantage estimates and suboptimal policy updates.

# Proximal Policy Optimization

- Entropy term

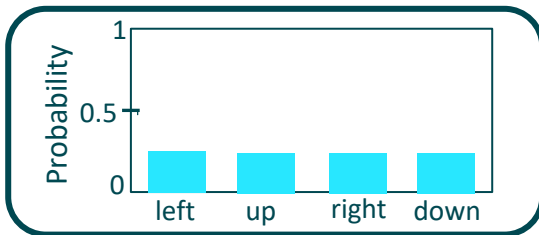
$$L_t^{PPO}(\theta) = \hat{E}_t[L_t^{CLIP}(\theta) - c_1 L_t^{VF}(\theta) + c_2 \boxed{S[\pi_\theta](s_t)}]$$

- Policy gradient algorithms may collapse into deterministic policies, so we add an entropy BONUS!
- Entropy measures the uncertainty of a distribution and is measured in bits.

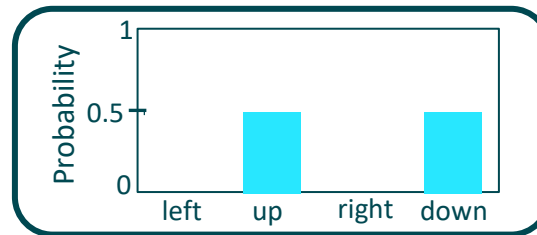
$$H(X) = - \sum_x p(x) \log_2 p(x)$$

- A higher entropy value indicates a more random distribution over actions, while a lower value suggests that the agent is becoming more predictable in its choices.

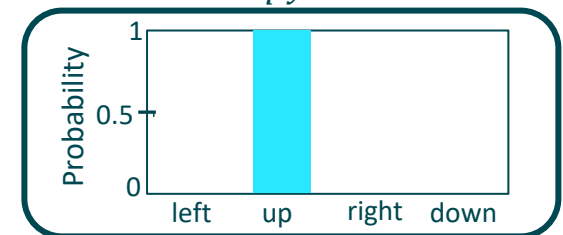
Entropy: 2 bits



Entropy: 1 bits



Entropy: 0 bits



# Proximal Policy Optimization

- Entropy term

- $c_2$  corresponds to the parameter ent\_coef

$$L_t^{PPO}(\theta) = \hat{E}_t[L_t^{CLIP}(\theta) - c_1 L_t^{VF}(\theta) + c_2 S[\pi_\theta](s_t)]$$

- It typically has a value of around 0.01.
  - High values of  $c_2$  will stimulate high entropy policies, to promote further exploration and discourage deterministic behaviour.
    - **High Entropy Coefficient:** When the entropy coefficient is set high, the agent is encouraged to explore a wider range of actions. This can lead to better generalization in complex environments, but it may also slow down convergence as the agent spends more time exploring.
    - **Low Entropy Coefficient:** On the other hand, a low entropy coefficient can lead to faster convergence as the agent focuses on taking actions that are known to be rewarding. However, this can sometimes lead to the agent finding suboptimal solutions too quickly, especially in environments that change a lot.

# Proximal Policy Optimization

- **Summary**

$$L_t^{PPO}(\theta) = \hat{E}_t[L_t^{CLIP}(\theta) - c_1 L_t^{VF}(\theta) + c_2 S[\pi_\theta](s_t)]$$

- **Policy improvement:** The clipped surrogate objective  $L_t^{CLIP}(\theta)$  ensures that the policy is improved in a stable manner.
- **Value Function Accuracy:** The value function error term  $-c_1 L_t^{VF}(\theta)$  ensures that the value function is learned accurately, which is critical for computing reliable benefit estimates.
- **Exploration:** The entropy term  $+c_2 S[\pi_\theta](s_t)$  encourages exploration by penalising low-entropy policies.

# Proximal Policy Optimization

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## Algorithm 1 PPO-Clip

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- 1: Input: initial policy parameters  $\theta_0$ , initial value function parameters  $\phi_0$
- 2: **for**  $k = 0, 1, 2, \dots$  **do**
- 3:   Collect set of trajectories  $\mathcal{D}_k = \{\tau_i\}$  by running policy  $\pi_k = \pi(\theta_k)$  in the environment.
- 4:   Compute rewards-to-go  $\hat{R}_t$ .
- 5:   Compute advantage estimates,  $\hat{A}_t$  (using any method of advantage estimation) based on the current value function  $V_{\phi_k}$ .
- 6:   Update the policy by maximizing the PPO-Clip objective:

$$\theta_{k+1} = \arg \max_{\theta} \frac{1}{|\mathcal{D}_k|T} \sum_{\tau \in \mathcal{D}_k} \sum_{t=0}^T \min \left( \frac{\pi_{\theta}(a_t|s_t)}{\pi_{\theta_k}(a_t|s_t)} A^{\pi_{\theta_k}}(s_t, a_t), \quad g(\epsilon, A^{\pi_{\theta_k}}(s_t, a_t)) \right),$$

typically via stochastic gradient ascent with Adam.

- 7:   Fit value function by regression on mean-squared error:

$$\phi_{k+1} = \arg \min_{\phi} \frac{1}{|\mathcal{D}_k|T} \sum_{\tau \in \mathcal{D}_k} \sum_{t=0}^T \left( V_{\phi}(s_t) - \hat{R}_t \right)^2,$$

typically via some gradient descent algorithm.

- 8: **end for**
-





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Eskerrik asko  
Muchas gracias  
Thank you

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